

Robustness report —

adata_Validation__pdex__20260608-202126

- dataset: /home/yangzhang/code/cell-eval/adata_Validation.h5ad
- DE backend: pdex (cell-level Wilcoxon; data normalize_total+log1p, is_log1p=True)
- pert_col / control: target_gene / non-targeting | sgrna_col: guide_id | replicate_col: batch
- scale: full, n_permutations=10, seed=0
- GLOBAL VERDICT: ⚠ WARN (validity gates tests 1-3 must PASS before interpreting tests 4-6)

> A validity gate WARNed — results are usable but read the flagged caveats (often a deflated/under-powered null) before interpreting tests 4-6.

Verification parameters & thresholds

parameter	value	source
de_method	pdex	config
allow_discrete / normalize_if_raw	False / True	config
fdr_threshold	0.05	config
lfc_threshold	0.1	config
lambda_gc_warn / fail	1.05 / 1.1	TEST_PLAN.md
lambda_gc deflation	WARN < 0.90	skill verdict notes
ks_p_uniform	WARN < 0.05	skill verdict notes
n_permutations	10	config
min_cells_per_group	20	config
seed	0	config
max_conditions	None	config
block_cols (stratify)	['batch']	config
test-3 separation	PASS / WARN > 2 / 1-2	TEST_PLAN.md
test-4 LOW-REPRODUCIBILITY flag	median LFC ρ < 0.2	TEST_PLAN.md
test-5 separation	PASS / WARN > 1.5 / 1.0-1.5	TEST_PLAN.md
test-6 recovery / direction	PASS > 0.5 / > 0.8	TEST_PLAN.md

Test summary

test	verdict	headline
test_1	⚠ WARN	Within-Condition Direct Split Null frac_sig=2.862e-06, mean_lfc=0.001775, mean_abs_lfc=0.1952, lambda_gc=0.7984, ks_p_uniform=3.957e-128, lambda_gc_sd=0.05431
test_2	⚠ WARN	Control-Control Split Null frac_sig=0, mean_lfc=0.0009492, mean_abs_lfc=0.1071, lambda_gc=1.005, ks_p_uniform=2.161e-14, lambda_gc_sd=0.03175
test_3	✅ PASS	Label Permutation Null true_signal=0.1475, perm_mean=0.007087, perm_sd=9.87e-05, separation_z=1423

| test_4 — Same-sgRNA Split Reproducibility | ☐ PASS | n_guides=54, median_lfc_spearman=0.2866, median_jaccard=0.2891, median_direction=0.8285 |

| test_5 — Same-Gene Independent sgRNA Reproducibility | ☐ FAIL | n_genes=4, n_same_pairs=4, same_gene_mean_rho=0.119, background_mean_rho=0.0458, separation_z=0.5281 |

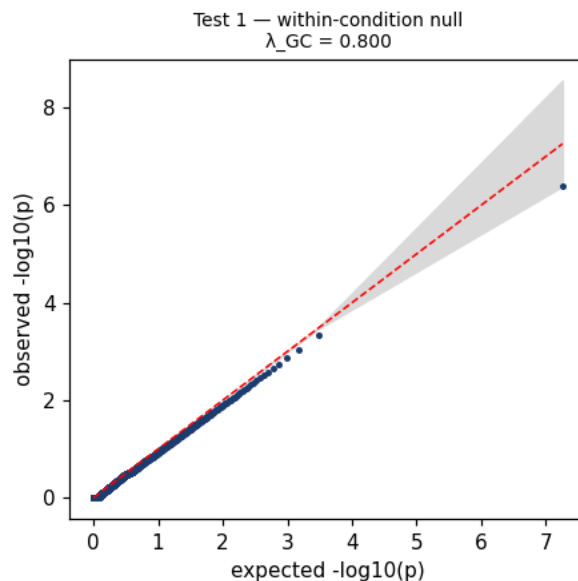
| test_6 — Target Gene Knockdown Recovery | ☐ PASS | n_targets=50, recovery_rate=1, direction_rate=1, median_pval_rank=5.531e-05, median_lfc_rank=0.1076 |

test_1 — Within-Condition Direct Split Null ⚠ WARN

- frac_sig = 2.8624193544954597e-06
- mean_lfc = 0.0017745508068358262
- mean_abs_lfc = 0.19523047652509917
- lambda_gc = 0.7984290521084172
- ks_p_uniform = 3.9574470333362476e-128
- lambda_gc_sd = 0.054310003160526775

p-values (verification): ks_p_uniform_mean = 3.957e-128, lambda_gc = 0.7984

- ⚠ $\lambda_{GC}=0.798 < 0.9$ (deflated null — under-powered)
- ⚠ ks_p_uniform=3.96e-128 < 0.05 (p-values not Uniform[0,1])
- ⚠ split-to-split SD(λ_{GC})=0.054 > 0.05 (residual confounding?)



Tables: see tables/test_1__*.csv

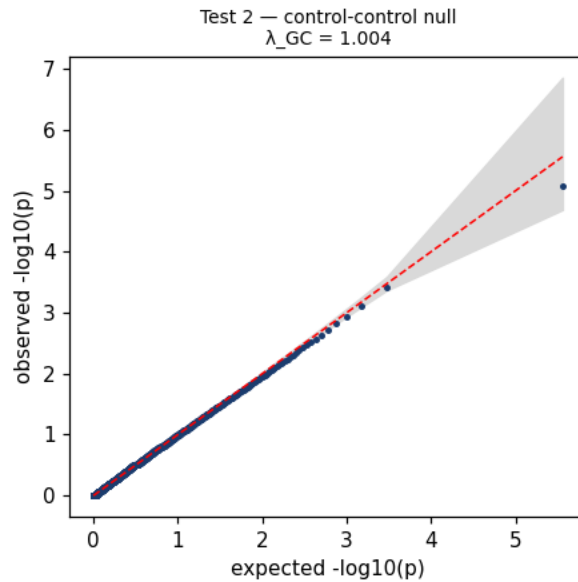
test_2 — Control-Control Split Null ⚠ WARN

- frac_sig = 0.0
- mean_lfc = 0.0009491504834339152
- mean_abs_lfc = 0.10714077880111354
- lambda_gc = 1.0049644906546602
- ks_p_uniform = 2.1610054394678046e-14

- $\lambda_{GC_sd} = 0.03174649284163869$

p-values (verification): $ks_p_uniform_mean = 2.161e-14$, $\lambda_{GC} = 1.005$

- $\Delta ks_p_uniform = 2.16e-14 < 0.05$ (p-values not Uniform[0,1])



Tables: see tables/test_2__*.csv

test_3 — Label Permutation Null ☒ PASS

- $true_signal = 0.14751216814159293$

- $perm_mean = 0.007086836283185842$

- $perm_sd = 9.869787552799573e-05$

- $separation_z = 1422.7796809929848$

p-values (verification): $perm_p = 0$, $separation_z = 1423$

Tables: see tables/test_3__*.csv

test_4 — Same-sgRNA Split Reproducibility ☒ PASS

- $n_guides = 54$

- $median_lfc_spearman = 0.2865950163645664$

- $median_jaccard = 0.2890805780399689$

- $median_direction = 0.8284725299069562$

Tables: see tables/test_4__*.csv

test_5 — Same-Gene Independent sgRNA Reproducibility ☐ FAIL

- $n_genes = 4$

- $n_same_pairs = 4$

- $same_gene_mean_rho = 0.11898355123049015$

- background_mean_rho = 0.0458045748388922

- separation_z = 0.5281370181808444

p-values (verification): separation_z = 0.5281

Tables: see tables/test_5__*.csv

test_6 — Target Gene Knockdown Recovery ☒ PASS

- n_targets = 50

- recovery_rate = 1.0

- direction_rate = 1.0

- median_pval_rank = 5.5309734513274336e-05

- median_lfc_rank = 0.10763274336283186

p-values (verification): median_padj_target = 0

Tables: see tables/test_6__*.csv

Power & sample-size analysis (TEST_PLAN.md §7)

§7.6 Realized parameters table

parameter	source	realized value
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$\alpha_{\text{empirical}}$	Tests 1-2 mean λ_{GC}	0.05
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λ_{GC} (mean of gates)	Tests 1-2	0.9017
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ϕ_{median} (approx)	control counts MoM	0.001
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δ_{floor}	true DE 10th pct LFC	0.1265
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δ_{typical}	true DE median LFC	0.2856
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δ_{ceiling}	true DE 90th pct LFC	1.127
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$\rho_{\text{reproducibility}}$	Test 4 median LFC ρ	0.2866
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$\delta_{\text{effective}} = \delta_{\text{typ}} \cdot \sqrt{\rho}$	derived	0.1529
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$n_{\text{required}}(\delta_{\text{typical}}, 80\%)$	power formula	464.4
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$n_{\text{required}}(\delta_{\text{effective}}, 80\%)$	power formula	1620
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$n_{\text{required}}(\delta_{\text{floor}}, 80\%)$	power formula	2369
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- α note: $\lambda_{\text{gc}} < 1$ (deflation) -> $\alpha_{\text{empirical}} = \text{fdr_threshold}$ (NOT inflated); investigate covariate overcorrection / under-powered null

- caveat: pdex is cell-level Wilcoxon (no NB dispersion fit); phi approximated by method-of-moments on control counts. Power is a planning estimate, not DESeq2-derived.

Appendix — embedded TEST_PLAN.md

Cell Metrics Robustness & Stability Test Plan

> Purpose. Validate that a cell-metrics evaluation pipeline produces calibrated, unbiased results before interpreting biological findings. Tests 1-3 are validity gates — all must pass before proceeding. Tests 4-6 are sensitivity

diagnostics — they characterise how much signal the data contains and set empirical ceilings for downstream metrics.

Inputs

Required

Field	Type	Description
adata	AnnData	Cells $\times$ genes count matrix
perturbation_col	str	obs column containing sgRNA / perturbation labels
control_label	str	Value in perturbation_col identifying control cells

Recommended

Field	Type	Description
target_gene_map	dict[str, str]	sgRNA $\rightarrow$ target gene name
batch_cols	list[str]	e.g. ['batch', 'donor', 'replicate', 'cell_type']
cell_cycle_col	str	obs column for cell-cycle phase
guide_efficacy	dict[str, float]	sgRNA $\rightarrow$ knockdown efficiency estimate
baseline_expression	dict[str, float]	Gene $\rightarrow$ baseline expression level

Parameters

Parameter	Default	Description
n_permutations	10	Resampling iterations for Tests 2 and 3
min_cells_per_group	20	Minimum cells per split arm; skip condition if not met
fdr_threshold	0.05	BH-adjusted p-value cutoff for significance calls
lfc_threshold	0.1	Minimum $ LFC $ for a gene to be called significant
lambda_gc_warn	1.05	λ_{GC} threshold above which a WARN is issued
lambda_gc_fail	1.10	λ_{GC} threshold above which a FAIL is issued

Shared Notation

Symbol	Definition
G	Number of genes tested
p(g)	Raw p-value for gene g
q(g)	BH-adjusted p-value for gene g
LFC(g)	Log ₂ fold change for gene g
S	Set of significant genes: $\{g : q(g) \leq \text{fdr_threshold} \text{ AND } LFC(g) \geq \text{lfc_threshold}\}$
F(x)	Empirical CDF of $\{p(g)\}_g$ evaluated at x
χ^2_{1-p}	Inverse CDF of the $\chi^2(1)$ distribution at probability p

Shared Formulas

All six tests report the following metrics where DE results are available.

Fraction Significant

,

$$\text{frac_sig} = |S| / G$$

,

Reported as the mean over conditions. Approximates the empirical false-positive rate under a true null. Expected value $\approx \text{fdr_threshold}$ when the null holds.

Mean and Mean Absolute LFC

,

$$\text{mean_lfc} = (1/G) \sum_g \text{LFC}(g)$$

$$\text{mean_abs_lfc} = (1/G) \sum_g |\text{LFC}(g)|$$

,

mean_lfc tests for directional bias; it can cancel to zero even with large symmetric LFCs.

mean_abs_lfc tests for magnitude inflation. Both must be reported together.

KS Uniformity Statistic

,

$$D = \sup_x |F(x) - x|$$

,

F is the empirical CDF of $\{p(g)\}_g$. The reported ks_p_uniform is the two-sided KS p-value against $\text{Uniform}[0,1]$. A small ks_p_uniform (< 0.05) indicates departure from the expected null p-value distribution.

Genomic Inflation Factor

,

$$\lambda_{GC} = \text{median}_g [\chi^2_{1-1}(1 - p(g))] / \chi^2_{1-1}(0.5)$$

,

where $\chi^2_{1-1}(0.5) = 0.4549$ (the median of a $\chi^2(1)$ distribution).

| λ_{GC} range | Interpretation |

|---|---|

| ≈ 1.00 | Well-calibrated |

| < 1.00 | Conservative / deflated |

| 1.00-1.05 | Acceptable |

| 1.05-1.10 | WARN — mild inflation |

| > 1.10 | FAIL — anti-conservative; do not proceed |

QQ-Plot Envelope

Plot sorted observed $-\log_{10} p_{(i)}$ ($i = 1 \dots G$) against expected $-\log_{10}((i - 0.5)/G)$.

The 95% pointwise confidence envelope uses the exact Beta order-statistic distribution:

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$$\text{lower visual bound} = -\log_{10}[B_{0.975}(i, G-i+1)]$$

upper visual bound = $-\log_{10}[B_{0.025}(i, G-i+1)]$

where $B_{\alpha}(\alpha, \beta)$ denotes the α -th quantile of the $Beta(\alpha, \beta)$ distribution. Note: the $-\log_{10}$ transform reverses the interval, so the lower Beta quantile ($B_{0.025}$) maps to the upper visual bound and vice versa. Points lying on $y = x$ indicate a well-calibrated test. This envelope is exact (not a normal approximation) because the i -th order statistic of $Uniform[0,1]$ follows exactly $Beta(i, G-i+1)$.

Wald Test Statistic

The recommended DE test for count data is the DESeq2 Wald test:

$$W = \beta / SE(\beta)$$

where β is the MLE $\log_2$ fold change estimated from the negative binomial model, and $SE(\beta)$ is derived from the Fisher Information at the MLE:

$$SE(\beta) = \sqrt{I(\beta)^{-1}}$$

w is the signal-to-noise ratio of the effect estimate: how many standard errors the estimated LFC sits from zero. Under H_0 ($\beta = 0$), asymptotic normality of the MLE gives $w \sim N(0, 1)$. The p-value is $p = 2\Phi(-|w|)$ where Φ is the standard normal CDF.

Why the Wald test: MLEs are asymptotically normally distributed, so dividing the estimate by its standard error yields a statistic with a known null distribution. DESeq2 additionally applies empirical Bayes shrinkage to both β and dispersions to stabilise estimates in the low-count, high-noise regime typical of RNA-seq.

Design formula: always include blocking variables as covariates:

$\sim$ replicate + batch + donor + split_label

This ensures the split coefficient is tested cleanly. Omitting covariates can produce false inflation even under a true null.

Alternative tests (use whichever matches the real analysis):

Test	Statistic	Notes
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edgeR QLF	F-statistic	Quasi-likelihood; more conservative; good for small n
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limma-voom	moderated t	Best for normalised log-CPM input
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Wilcoxon rank-sum	W	Non-parametric; ignores replicate structure; use only if others unavailable
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> Critical: the null split tests must use the same DE method as the real analysis. They are a calibration check on the pipeline, not a standalone test.

Test 1 — Within-Condition Direct Split Null

Question

If cells from the same condition are split into two balanced groups, does the pipeline detect little to no difference between them?

Design

For each perturbation condition with $\geq 2 \times \text{min_cells_per_group}$ cells:

1. Perform $n_{\text{permutations}}$ independent stratified splits of the condition's cells into arms A and B, stratified by all available covariates in `batch_cols` plus `cell_cycle_col`.
2. For each split, run DE: A vs B, with covariates in the design formula.
3. Compute shared metrics per split: `frac_sig`, `mean_lfc`, `mean_abs_lfc`, `ks_p_uniform`, λ_{GC} , QQ-plot envelope.
4. Aggregate metrics as mean $\pm$ SD across splits within each condition, then across all conditions.

> Why $n_{\text{permutations}}$ here: a single split can be unlucky — one draw may by chance separate a latent covariate. Averaging over multiple independent stratified splits gives a stable estimate of null behaviour and exposes split-to-split variance that would indicate residual confounding.

Test Statistic

Wald statistic $W = \beta_{\text{A}} / \text{SE}(\beta_{\text{A}})$ from the negative binomial model, where β_{A} is the A-vs-B LFC. Under H_0 , $W \sim N(0, 1)$ and p-values are Uniform[0,1]. Report the mean W distribution pooled across all splits and conditions.

Expected Behaviour

Metric	Expected
frac_sig	$\leq \text{fdr_threshold}$ ($\approx 5\%$)
mean_lfc	≈ 0.0
mean_abs_lfc	Near zero; low relative to real analysis
ks_p_uniform	> 0.05 (fail to reject uniformity)
λ_{GC}	≈ 1.00
Split-to-split SD of λ_{GC}	Low (< 0.05); high SD indicates residual confounding
QQ-plot	Points hug $y = x$ diagonal within envelope; stable across splits

Verdict Rules

Condition	Verdict
All metrics within expected range	PASS
$\lambda_{\text{GC}} \in (1.05, 1.10]$ OR <code>frac_sig</code> modestly elevated	WARN
$\lambda_{\text{GC}} > 1.10$ OR <code>frac_sig</code> $\gg \text{fdr_threshold}$ OR QQ grossly inflated	FAIL

Failure Diagnostics

Pattern	Likely Cause
p-value histogram spikes near 0	Unabsorbed batch structure leaking into split
LFCs skewed in one direction	Normalization bias or unbalanced split
QQ inflation (points above diagonal)	Missing covariates; overdispersion underestimated
QQ deflation (points below diagonal)	Overcorrection; too many covariates

Test 2 — Control-Control Split Null

Question

Do metrics remain near null when pseudo-perturbations are created by splitting control cells?

Design

1. Take all cells where `perturbation_col == control_label`.
2. Perform `n_permutations` independent stratified splits into groups A and B.
3. In each split, treat group A as a pseudo-perturbation and group B as the reference.
4. Run the full cell-metrics workflow for each split.
5. Aggregate metrics: mean $\pm$ SD across splits.

Test Statistic

Same Wald test as Test 1. Additionally compute all downstream cell-metrics outputs (overlap, AUC, delta correlations) to verify they also collapse to null.

Expected Behaviour

Metric Expected
--- ---
<code>frac_sig</code> $\approx$ <code>fdr_threshold</code>
<code>mean_lfc</code> , <code>mean_abs_lfc</code> Near zero
<code>ks_p_uniform</code> > 0.05
λ_{GC} ≈ 1.00
Split-to-split SD of λ_{GC} Low (< 0.05)
QQ-plot Points hug $y = x$ diagonal within envelope; consistent across splits
DEG overlap / precision / AUC Near chance level
Delta correlations Near zero
AnnData-level deltas Near zero

Verdict Rules

Same thresholds as Test 1, applied to the mean over splits. Additionally: if cell-metrics scores (overlap, AUC) are substantially above chance across splits, verdict is FAIL.

Test 3 — Label Permutation Null

Question

Do metrics collapse when perturbation labels are broken?

Design

1. Compute all metrics under true perturbation labels $\rightarrow$ `true_metrics`.
2. For each of `n_permutations` iterations:
 - a. Shuffle `perturbation_col` values within each stratum defined by `batch_cols`.
 - b. Recompute DE and all cell-metrics $\rightarrow$ `perm_metrics[i]`.
3. Compute separation score between true and permuted distributions.

Separation Score

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$$\text{separation} = (\text{true_metric} - \text{mean}(\text{perm_metrics})) / \text{SD}(\text{perm_metrics})$$

,

This is equivalent to a z-score of the true metric relative to the permutation null. Also compute an empirical p-value:

$\text{perm_p} = |\{i : \text{perm_metrics}[i] \geq \text{true_metric}\}| / n_{\text{permutations}}$

Expected Behaviour

Metric	Expected
Permuted frac_sig , λ_{GC}	Indistinguishable from Test 1/2 null
Separation score for strong metrics	> 2.0 (true signal clearly outside null)
perm_p for strong metrics	< 0.05

Verdict Rules

Condition	Verdict
True metrics sit clearly outside permutation null ($\text{separation} > 2$)	PASS
Separation borderline ($1 < \text{separation} \leq 2$)	WARN
True metrics indistinguishable from permuted ($\text{separation} \leq 1$)	FAIL — metrics cannot detect real signal

Test 4 — Same-sgRNA Split Reproducibility

Question

If the same sgRNA's cells are split into two groups and each is compared to control, are the resulting DE signatures reproducible?

Design

For each sgRNA with $\geq 2 \times \text{min_cells_per_group}$ perturbed cells:

- Split perturbed cells into arms A and B (stratified by `batch_cols`).
- Run DE: A vs control $\rightarrow \text{de_A}$.
- Run DE: B vs control $\rightarrow \text{de_B}$.
- Compare de_A and de_B using reproducibility metrics below.

Reproducibility Metrics

Metric	Formula	Interpretation
LFC correlation	$\rho = \text{Spearman}(\text{LFC_A}, \text{LFC_B})$ over all G genes	Overall signature agreement
DEG overlap	$\text{Jaccard}(S_A, S_B) = \frac{ S_A \cap S_B }{ S_A \cup S_B }$	Significant gene set overlap
Directional agreement	$(1/G) \sum_g \mathbb{1}[\text{sign}(\text{LFC_A}(g)) = \text{sign}(\text{LFC_B}(g))]$	Fraction of genes agreeing in direction
AUC recovery	AUROC of S_A ranks predicting S_B membership	Rank-based reproducibility

Expected Behaviour

Perturbation strength	LFC correlation	DEG Jaccard	Directional agreement
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Strong	> 0.6	> 0.3	> 0.7
Moderate	0.3-0.6	0.1-0.3	0.6-0.7
Weak	< 0.3	< 0.1	$\approx$ 0.5 (chance)

> Note: This test sets the empirical ceiling for all downstream evaluation metrics. A perturbation with low reproducibility here cannot be expected to score well on cross-model or cross-condition comparisons.

Verdict Rules

Reproducibility is not a pass/fail gate but an empirical characterisation. Flag perturbations with LFC correlation < 0.2 as LOW REPRODUCIBILITY and exclude them from sensitivity benchmarks.

Test 5 — Same-Gene Independent sgRNA Reproducibility

Question

Do independent sgRNAs targeting the same gene recover similar perturbation signatures?

Design

For each target gene G with ≥ 2 sgRNAs:

1. Compute DE for each sgRNA versus control.
2. For all same-gene guide pairs (i, j): compute reproducibility metrics from Test 4.
3. For matched unrelated guide pairs (random pairs from different target genes): compute the same metrics $\rightarrow$ background distribution.
4. Optionally run leave-one-guide-out (LOGO) analysis:
 - Reference signature: consensus of all sgRNAs targeting G except guide i.
 - Query signature: guide i alone.
 - Compare LOGO scores against same-gene pair scores.

Separation Score (same-gene vs background)

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$$\text{separation} = (\text{mean_same_gene_metric} - \text{mean_background_metric}) / \text{SD}(\text{background_metric})$$

,

Expected Behaviour

Metric	Expected
Same-gene LFC correlation	Substantially higher than background
Same-gene DEG Jaccard	Substantially higher than background
Separation score	> 1.5 for reliable target gene recovery

Verdict Rules

Condition	Verdict
Same-gene pairs score clearly above background	PASS
Marginal separation ($1.0 < \text{separation} \leq 1.5$)	WARN — possible off-target effects or weak perturbation
Same-gene pairs indistinguishable from background	FAIL — guides may be off-target or inefficacious

Test 6 — Target Gene Knockdown Recovery

Question

Is the target gene itself detected as differentially expressed in the expected direction?

Design

For each sgRNA with a known target gene G (from target_gene_map):

1. Run DE: perturbed cells vs control.
2. Extract result for gene G.
3. Record the following per-sgRNA outputs.

Per-sgRNA Output Fields

Field	Formula / Source	Description
lfc_target	LFC(G) from DE	Log ₂ fold change of target gene
padj_target	q(G) from DE	BH-adjusted p-value of target gene
pval_rank	rank(p(g))[G] / G	Percentile rank of target gene by raw p-value (lower = more significant)
lfc_rank	rank(LFC(g))[G] / G	Percentile rank by absolute LFC
correct_direction	LFC(G) < 0 for CRISPRi/KO	Whether LFC is in the expected direction
detected	q(G) ≤ fdr_threshold	Binary: is target gene significant?

Aggregate Recovery Metrics

Computed over all sgRNAs with a mapped target gene:

```
recovery_rate = |{sgRNAs : detected = True}| / N_sgrnas
direction_rate = |{sgRNAs : correct_direction = True}| / N_sgrnas
median_pval_rank = median over sgRNAs of pval_rank
median_lfc_rank = median over sgRNAs of lfc_rank
```

Expected Behaviour

Metric	Expected (good data)	Warning threshold
recovery_rate	> 0.5	< 0.2
direction_rate	> 0.8 (CRISPRi/KO)	< 0.6
median_pval_rank	< 0.10 (top 10% by p-value)	> 0.25
median_lfc_rank	< 0.15	> 0.30

> Note for CRISPR KO: reduced transcript abundance is expected for many genes but is not guaranteed due to nonsense-mediated decay variability. direction_rate thresholds may need adjustment based on the gene and perturbation system.

Verdict Rules

Condition	Verdict
recovery_rate > 0.5 and direction_rate > 0.8	PASS

| recovery_rate $\in$ [0.2, 0.5] or direction_rate $\in$ [0.6, 0.8] | WARN — check guide efficacy estimates |
| recovery_rate < 0.2 or direction_rate < 0.6 | FAIL — poor guide efficacy or expression artefact |

Output Schema

`python

@dataclass

class TestResult:

verdict: Literal["PASS", "WARN", "FAIL", "SKIP"]

metrics: dict[str, float] # all numeric outputs for this test

flags: list[str] # human-readable failure reasons

details: dict # per-condition or per-sgRNA breakdown

@dataclass

class RobustnessReport:

test1: TestResult # Within-condition null

test2: TestResult # Control-control null

test3: TestResult # Label permutation

test4: TestResult # Same-sgRNA reproducibility

test5: TestResult # Cross-sgRNA reproducibility

test6: TestResult # Target knockdown recovery

global_verdict: Literal["PASS", "WARN", "FAIL"]

blocking_flags: list[str] # reasons for FAIL at global level

info_flags: list[str] # non-blocking WARNs

`

Global Verdict Logic

| Rule | Global Verdict |

|---|---|

| Tests 1, 2, 3 all PASS | Proceed to Tests 4–6 |

| Any of Tests 1–3 is FAIL | FAIL — do not interpret real results; fix pipeline first |

| Tests 1–3 PASS but Test 6 WARN | WARN — results valid but guide efficacy may limit sensitivity |

| Tests 4–5 show no separation from background | WARN — empirical ceiling is low; downstream metrics may not be informative |

| Tests 4–5 FAIL and Tests 1–3 also FAIL | FAIL |

Implementation Notes

1. Shared split function. Tests 1, 2, and 4 all require stratified A/B splits. Implement a single stratified_split(cells, stratify_cols, min_per_group) utility used by all three.

2. Shared reproducibility function. Tests 4 and 5 both compare two DE result objects. Implement a single compare_signatures(de_A, de_B) -> ReproducibilityMetrics used by both.

3. SKIP conditions. If a condition has fewer than $2 \times \text{min_cells_per_group}$ cells, mark that condition as SKIP and exclude from aggregation. If $> 50\%$ of conditions are SKIP, escalate to WARN.
4. Consistency requirement. All six tests must use the same DE function as the real analysis pipeline. Pass the DE function as a callable argument to each test runner.
5. Permutation seed. Set a fixed random seed for all stratified splits and permutations to ensure reproducibility of the report.

Section 7 — From Realized Tests to a Real DE Analysis

Once all six robustness tests have been run, their outputs collectively define the statistical parameters for the real analysis. This section specifies exactly how each realized result maps to a p-value threshold, effect size estimate, power calculation, and required sample size.

7.1 Calibrated α from Tests 1 and 2

The nominal FDR threshold ($\text{fdr_threshold} = 0.05$) assumes a well-calibrated test. The realized null tests provide an empirical α that corrects for any residual inflation.

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$$\alpha_{\text{empirical}} = \text{fdr_threshold} / \lambda_{\text{GC}}$$

,

where λ_{GC} is the mean genomic inflation factor from Tests 1 and 2.

λ_{GC} (realized)	$\alpha_{\text{empirical}}$ (if $\text{fdr_threshold} = 0.05$)	Action
1.00	0.050	Use nominal threshold unchanged
1.05	0.048	Minor adjustment; use nominal
1.10	0.045	Apply corrected threshold
1.20	0.042	Apply corrected threshold; investigate cause
> 1.10	—	FAIL — do not run real analysis until resolved

|---|---|---|

| 1.00 | 0.050 | Use nominal threshold unchanged |

| 1.05 | 0.048 | Minor adjustment; use nominal |

| 1.10 | 0.045 | Apply corrected threshold |

| 1.20 | 0.042 | Apply corrected threshold; investigate cause |

| > 1.10 | — | FAIL — do not run real analysis until resolved |

The real DE analysis should use $\alpha_{\text{empirical}}$ as the BH input threshold rather than the raw nominal level. If $\lambda_{\text{GC}} < 1$ (deflation), do not inflate α — instead flag the deflation and investigate covariate overcorrection.

p-value interpretation rule for the real analysis:

A gene is called significant in the real analysis if and only if:

,

$$q(g) \leq \alpha_{\text{empirical}} \text{ AND } |\text{LFC}(g)| \geq \text{lfc_threshold}$$

,

where lfc_threshold is informed by Test 4 (see Section 7.3).

7.2 Empirical Effect Size from Tests 4 and 5

Tests 4 and 5 yield a distribution of LFC correlations and DEG overlaps across sgRNAs. These define three effect size tiers used in power calculations.

From Test 4 (same-sgRNA split):

,

lfc_sd_signal = SD of LFC(g) across significant genes, averaged over sgRNAs

$lfc_ceiling$ = median LFC of top-decile significant genes across sgRNAs

lfc_floor = 10th percentile of $|LFC|$ among detected DEGs

,

From Test 6 (target knockdown):

,

Δ_{target} = median $|lfc_target|$ across sgRNAs where detected = True

,

Effect size tiers:

Tier	Definition	Use
------	------------	-----

Strong	$ LFC \geq lfc_ceiling$	Upper bound on detectable effect; sets optimistic power
--------	---------------------------	---

Typical	$ LFC = \text{median}(LFC_detected)$ from Test 4	Primary effect size for power calculation
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Minimal detectable	$ LFC = lfc_floor$	Conservative lower bound; sets required n
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For the real analysis, use the typical effect size as the primary estimand. Report power at all three tiers.

7.3 Refined $lfc_threshold$ from Tests 4 and 6

The initial $lfc_threshold = 0.1$ is a prior. Tests 4 and 6 provide data to refine it.

,

$lfc_threshold_realized = \max(lfc_floor, lfc_threshold_prior)$

,

If lfc_floor from Test 4 is substantially above 0.1 (e.g. 0.3), raising the threshold reduces false positives without sacrificing real signal. If lfc_floor is below 0.1, the prior is conservative and can be relaxed only if Test 1 and 2 are cleanly calibrated.

7.4 Power Calculation

Power for detecting a gene with true LFC δ at the calibrated threshold $\alpha_{empirical}$, given n cells per group:

Wald test power (negative binomial approximation):

,

$SE(\beta) \approx \sqrt{(1/\mu_A + 1/\mu_B + \phi) / n}$

$W_{ncp} = \delta / SE(\beta)$

Power = $\Phi(W_{ncp} - z_{\alpha/2}) + \Phi(-W_{ncp} - z_{\alpha/2})$

,

where:

Symbol	Definition
--------	------------

δ	True LFC (use typical effect size from Section 7.2)
----------	---

| μ_A, μ_B | Mean count in each group (use baseline expression from baseline_expression or empirical mean from adata) |

| ϕ | Dispersion estimate from DESeq2 fit on the null split data |

| n | Cells per group |

| $z_{\{\alpha/2\}}$ | Normal quantile at $\alpha_{\text{empirical}} / 2$ (two-tailed) |

| Φ | Standard normal CDF |

Dispersion ϕ from robustness tests: use the median fitted dispersion from the Test 1 or Test 2 DESeq2 run. This is data-derived and more accurate than a prior.

Multiple testing correction on power: since G genes are tested, the effective per-gene α after BH correction is approximately:

$$\alpha_{\text{per_gene}} \approx \alpha_{\text{empirical}} \times (G_{\text{sig}} / G)$$

where G_{sig} is the expected number of true positives (estimated from Test 4 recovery rate $\times$ G). Substitute $\alpha_{\text{per_gene}}$ for $\alpha_{\text{empirical}}$ in the power formula for a conservative estimate.

7.5 Required Sample Size

Invert the power formula to solve for n given a target power (typically 0.80):

$$n_{\text{required}} = \left[(z_{\{\alpha/2\}} + z_{\{1-\beta\}}) / (\delta / \sqrt{1/\mu_A + 1/\mu_B + \phi}) \right]^2$$

where $z_{\{1-\beta\}}$ is the normal quantile at target power (0.84 for 80%, 1.28 for 90%).

Compute n_{required} at each effect size tier from Section 7.2:

Tier	δ	n_{required} (80% power)	Interpretation
Strong	lfc_ceiling	Lowest n	Optimistic; achievable for top-effect genes
Typical	median LFC	Middle n	Primary planning target
Minimal detectable	lfc_floor	Highest n	Conservative; needed to detect weakest real signals

Adjustment for reproducibility ceiling (Test 4):

If the same-sgRNA LFC correlation from Test 4 is $\rho < 1$, the effective detectable LFC is attenuated:

$$\delta_{\text{effective}} = \delta \times \sqrt{\rho}$$

Substitute $\delta_{\text{effective}}$ into n_{required} . This accounts for the fact that no single comparison can recover more signal than the data's internal reproducibility allows.

7.6 Summary: Realized Parameters Table

After running all six tests, populate this table. It becomes the statistical specification for the real analysis.

Parameter	Source	Realized Value
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|---|---|---|

| $\alpha_{\text{empirical}}$ | Tests 1-2: mean λ_{GC} | $\text{fdr_threshold} / \lambda_{\text{GC}}$ |

| $\text{lfc_threshold_realized}$ | Tests 4, 6: lfc_floor | $\max(\text{lfc_floor}, 0.1)$ |

| ϕ_{median} | Test 1 or 2: DESeq2 dispersion fit | empirical |

| δ_{typical} | Test 4: median detected LFC | empirical |

| δ_{ceiling} | Test 4: top-decile LFC | empirical |

| δ_{floor} | Test 4: 10th percentile detected LFC | empirical |

| $\rho_{\text{reproducibility}}$ | Test 4: median same-sgRNA LFC correlation | empirical |

| $\delta_{\text{effective}}$ | $\delta_{\text{typical}} \times \sqrt{\rho}$ | derived |

| $n_{\text{required_typical}}$ | Power formula at δ_{typical} , 80% power | derived |

| $n_{\text{required_conservative}}$ | Power formula at δ_{floor} , 80% power | derived |

| recovery_rate | Test 6 | empirical |

| direction_rate | Test 6 | empirical |

Decision rule for the real analysis:

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IF $\text{global_verdict} == \text{PASS}$:

use $\alpha_{\text{empirical}}$, $\text{lfc_threshold_realized}$, $n_{\text{required_typical}}$

report power at all three δ tiers

ELIF $\text{global_verdict} == \text{WARN}$:

use $\alpha_{\text{empirical}}$ with additional Bonferroni factor of 2

flag that power estimates may be optimistic

ELIF $\text{global_verdict} == \text{FAIL}$:

STOP — do not run real analysis

return blocking_flags for diagnosis

,